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876

Development of zero

Although mathematicians had worked out problems involving the use of zero, there was no symbol for it before the 9th century. An inscription dated 876 from Gwalior, India, contains the first known use of a symbol for zero in describing the dimensions of a garden. Its appearance allowed the development of a full decimal system for numbers.

“We should not be ashamed to acknowledge the truth or to acquire it, wherever it comes from.”

al-Kindi, Arabic mathematician and philosopher, c 800–873

Movable plates adjust the astrolabe's alignment and help the user to calculate the positions of astronomical objects.

Star pointer shows the position of a particular star.



Mapping the sky

A device called an astrolabe helped ancient astronomers to calculate the positions of stars and other objects in the sky. Around 920, an Arab astronomer, al-Battani, worked out the complicated calculations needed to use the astrolabe.

Ring represents the pathway of the Sun through the sky.

c 854–925

AL-RAZI

Born in Rayy (now in Iran), al-Razi was one of the Arabic world's greatest physicians. He was the first to describe hayfever and the symptoms of smallpox. Unlike most doctors of the time, he did not support the theory that an incorrect balance of body fluids known as “humors” affected health.



al-Razi with an assistant in his laboratory

Classifying elements

Interested in alchemy (medieval chemistry), al-Razi devised a system for classifying elements. He divided substances into spirits, metals, and minerals, noting what happened to each when it was heated or subjected to chemical processes.

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